

DIGITAL FORENSICS

Forensic Lifecycle of a File Created Using CLI Stage 2

Collaboration Note

This report is part of a collaborative digital forensics series between Julian Derry and Maxwell B. Antwi (TraceHunter) documenting the forensic lifecycle of a file.

The objective is to demonstrate how the same outcome, file creation, can originate from two different user workflows and how each leaves a distinct evidential trail.

- Maxwell B. Antwi: GUI-based file creation and graphical user interaction artifacts.
- Julian Derry: CLI-based file creation and corroboration using PowerShell, ConsoleHost History, \$MFT, \$UsnJrnl, and \$LogFile.

Together, the series shows how GUI and CLI workflows converge on the same underlying NTFS evidence while producing different contextual artifacts.

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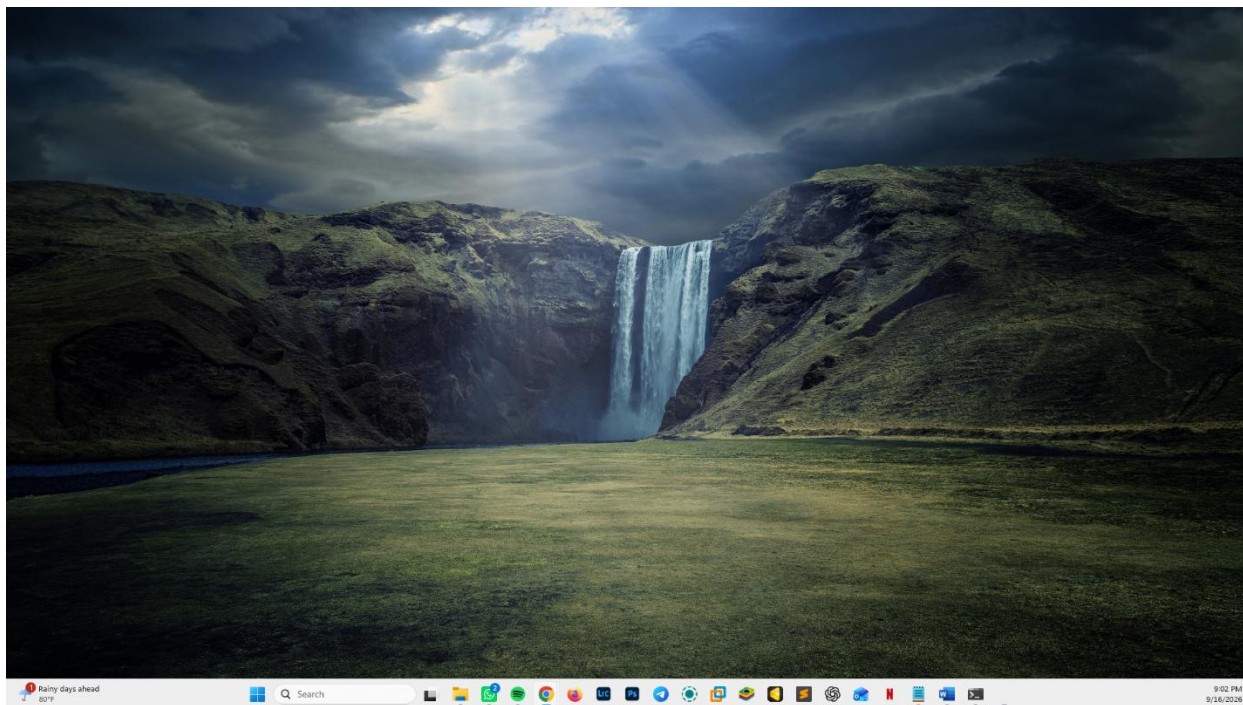
Case Reference No.	DFIR-2026-FFLC
Investigator	Julian Derry
Date of Analysis	September 16, 2026
Prepared For	Public
Platform	Windows
Tools	GKAPE, Eric Zimmerman Tools

Step 1. Objective

Investigation Question

Can I prove that a file was created from the Command Line Interface (CLI), and distinguish that workflow from GUI-based file creation?

This report documents the **CLI path** of a file's forensic lifecycle as part of a collaborative series. The GUI workflow is documented separately; this investigation focuses exclusively on evidence generated through PowerShell.

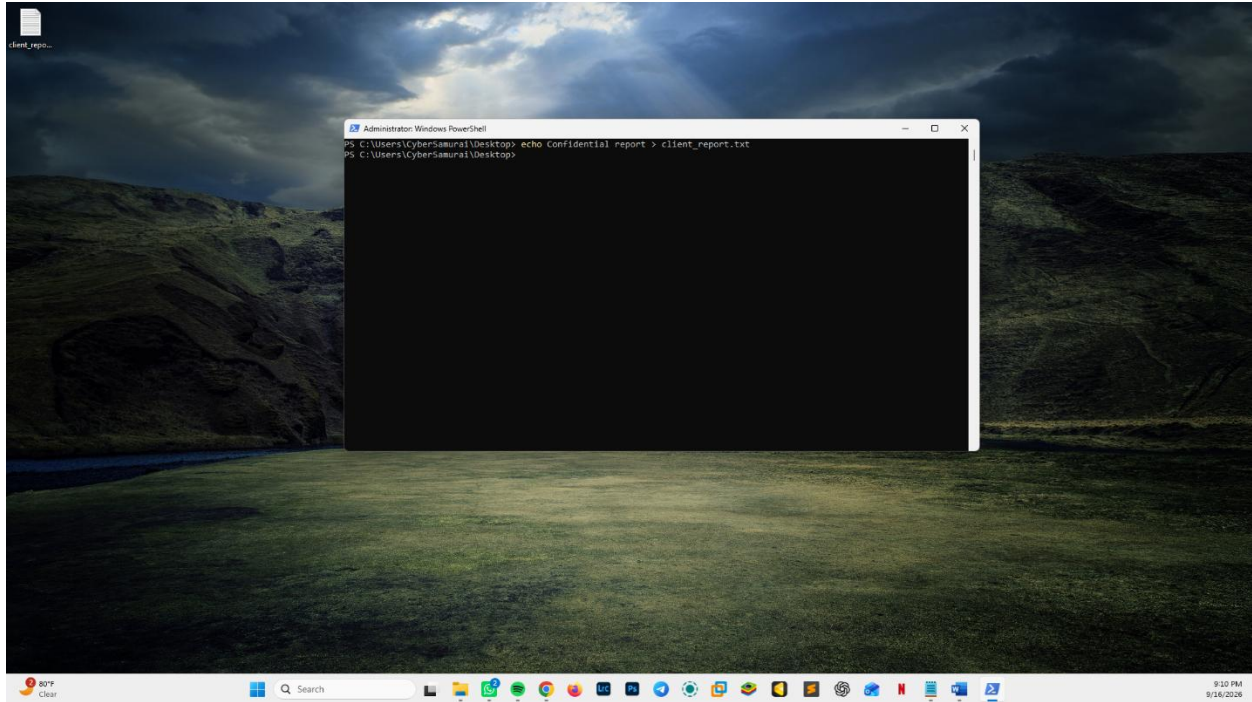


Step 2. Build the lab

A Windows 11 virtual machine was used for this experiment.

A folder structure was created using PowerShell only, without interacting with File Explorer. This ensured the resulting artifacts represented a genuine CLI workflow.

The file was created with the following command: `echo Confidential report > client_report.txt`



Creation Details

Item	Value	Purpose
System time	9/16/2026 9:10 PM	Timeline anchor
Full file path	\Users\CyberSamurai\Desktop	Location of birth
Command used	echo Confidential report > client_report.txt	User action
User account	CyberSamurai	Context

Step 4. Acquire evidence

Evidence was acquired using **gKAPE**.

Rather than collecting individual artifacts manually, the **KapeTriage** target was selected to preserve the core Windows forensic artifacts required for timeline reconstruction.

File System Artifacts

The following NTFS artifacts were collected:

- \$MFT
- \$LogFile
- \$UsnJrnl

These artifacts establish the physical creation of the file within the NTFS file system.

Event Logs

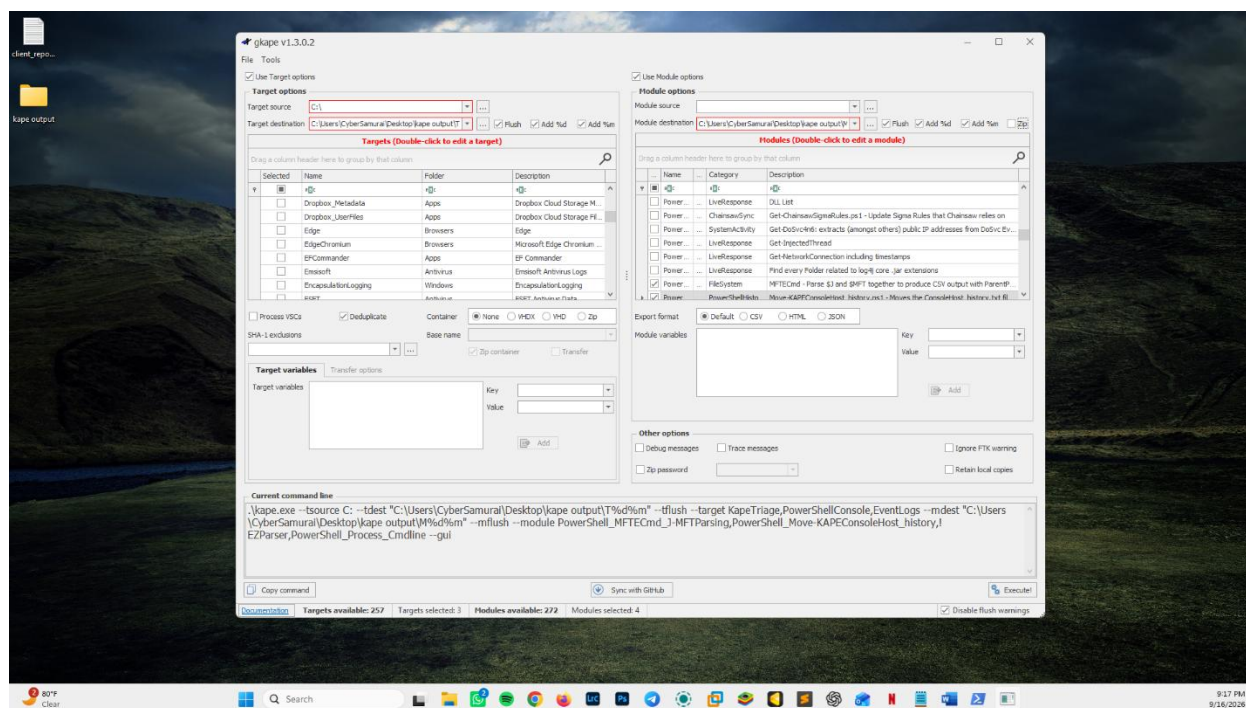
The following logs were collected:

- Security.evtx
- System.evtx
- Microsoft-Windows-PowerShell/Operational.evtx

For redundancy, the **EventLogs** target was acquired separately in addition to KapeTriage.

PowerShell History

The **ConsoleHost_history.txt** artifact was collected to preserve the commands executed during the PowerShell session.



File system

Instead of collecting the system files individually, I selected KapeTriage, which collects all the system files. This is to establish physical creation.

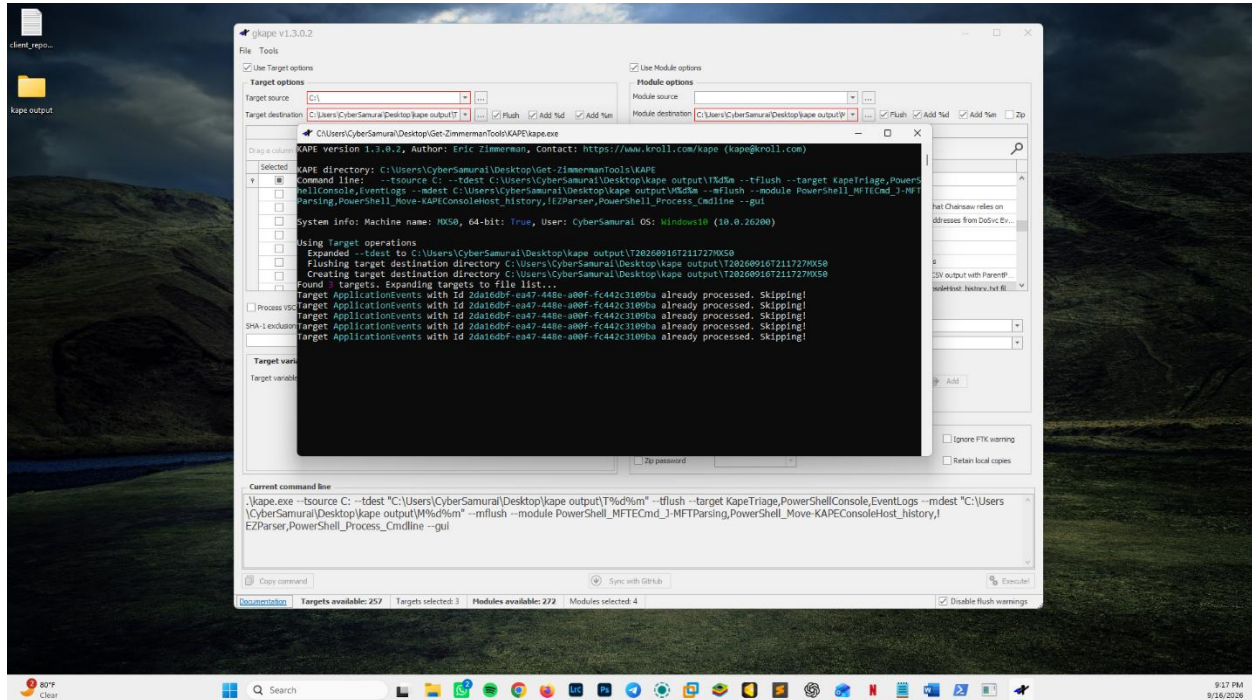
Registry

KapeTriage also collects the needed files such as UserAssit, RecentDocs and ShellBags. Little or no activity if the workflow stayed in CLI. That absence becomes meaningful.

Step 4. Parse the Artifacts

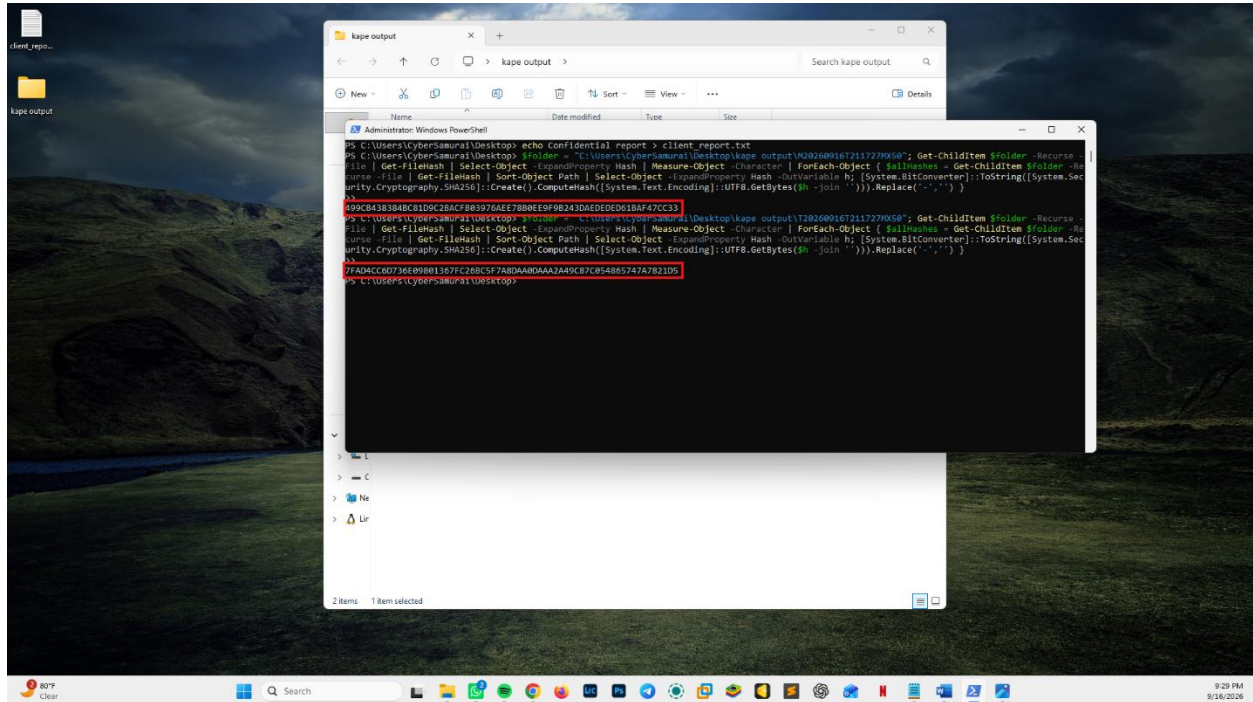
All collected targets were parsed using Eric Zimmerman's EZParser modules.

The parsed output produced structured CSV files for timeline correlation across NTFS, Registry, and Event Log artifacts.



Step 4.1 Evidence Integrity

Before analysis, hash values were calculated for the parsed output folders to preserve evidential integrity throughout the examination.



Step 5. Correlate the Timeline

Independent artifacts were correlated to determine whether they described the same creation event.

Time	Artifact	Interpretation
9:10:19 PM 9/16/2026	\$MFT	File record allocated
9:10:19 PM 9/16/2026	USN Journal	NTFS records file creation
9:10:19 PM 9/16/2026	\$LogFile	Creation transaction committed
9:10:19 PM 9/16/2026	ConsoleHost_history.txt	User issued creation command

The convergence of these independent artifacts corroborates a single file creation event originating from the CLI workflow.

Step 6. Artifact Findings

Parsed \$MFT

The Master File Table confirms the allocation of client_report.txt and records its creation timestamps.

DIGITAL FORENSICS & INCIDENT RESPONSE FINAL REPORT

mft_parsed.csv - Excel															Julian D				D					
File Home Insert Page Layout Formulas Data Review View Help Tell me what you want to do															Share									
Cut Copy Paste Format Painter Clipboard Font Alignment Number Conditional Formatting Table Styles Cells AutoSum Fill Sort & Find Filter Select Add-ins																								
Calibri 11 A A A Wrap Text General Normal Bad Good Neutral Calculation Check Cell Explanatory Input																								
A105764 104464																								
1	EntryN	Sequel	InUse	Parent	Parent	ParentPath	FileName	Extens	FileSize	Refere	Rep	IsDirect	HasAds	IsAds	SkFN	uSec2e	Copied	Siflags	NameT	Created	Created	LastModified	DwL	LastModi
42631	40463	77	TRUE	5704	105	Users\CyberSamurai\Desktop	client_report.txt	.txt	46	1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	Archive	Windows	9/16/26 9:10 PM	9/16/26 9:10 PM			
42749	42302	245	TRUE	13821	20	Users\CyberSamurai\AppData\Local\Software\Microsoft\Windows\CurrentVersion\SettingsCon	SettingsCon.txt	.txt	10672	1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	Archive	Windows	9/16/26 9:07 PM	9/16/26 9:07 PM			
67038	66627	187	TRUE	7663	17	Filecycle Bin\5-1-5-21-2539662315-58000UK.txt	58000UK.txt	.txt	240	1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	Archive	DosWind	9/16/26 9:12 PM	9/16/26 9:12 PM			
98164	96817	181	TRUE	504612	28	Users\CyberSamurai\Desktop\Get log.txt	Get log.txt	.txt	146	1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	Archive	DosWind	9/16/26 9:36 PM	9/16/26 9:37 PM			9/16/2
105764	104464	464	TRUE	1207	301	Windows\System32	client_report.txt	.txt	46	1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	Archive	Windows	9/16/26 9:07 PM	9/16/26 9:07 PM			
179176	175417	208	TRUE	174218	28	Users\CyberSamurai\AppData\Local\2026-09-16.txt	2026-09-16.txt	.txt	20395	1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	Archive	Windows	9/16/26 12:40 AM	9/16/26 9:18 PM	9/16/26		
244693	239754	211	TRUE	245680	93	Users\CyberSamurai\Desktop\Get 2026-09-16T21_17_27_24753.txt	24753.txt	.txt	72328	1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	Archive	Windows	9/16/26 9:19 PM	9/16/26 9:19 PM	9/16/26		
249773	244178	117	TRUE	7663	17	Filecycle Bin\5-1-5-21-2539662315-58000UK.txt	58000UK.txt	.txt	240	1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	Archive	DosWind	9/16/26 9:12 PM	9/16/26 9:12 PM	9/16/26		
250823	245482	219	TRUE	138318	40	Users\CyberSamurai\AppData\Local\ipc.txt	ipc.txt	.txt	0	1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	Archive	DosWind	9/16/26 9:20 PM	9/16/26 9:20 PM	9/16/26		

Parsed \$UsnJrnl

The USN Journal records the file creation event, providing transactional evidence that the file was created on disk.

j_parsed.csv - Excel															Autosave		Share			
File Home Insert Page Layout Formulas Data Review View Help Tell me what you want to do															Fill		Sort & Find		Add-ins	
Clipboard Font Paragraph Styles															Conditional Formatting		Calculation		Editing	
A117941 client_report.txt															Check Cell		Explanatory...		Add-ins	
															Cells					
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q			
1	Name	Extens	EntryNumber	SequenceNumber	ParentEntryN	ParentSeque	Parent	UpdateSequenceN	UpdateTimestamp	UpdateReasons	FileAttributes	Offset	Source							
114290	WER.5c1587a6-4442-46b9-a.txt	.txt	212143	74	888460	4	4217294078	9/16/26 6:30 PM	FileCreate	Archive	NotContentIndexed	25625056	C:\Users\CyberSamurai\Desktop\Si							
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116127	WER.2f6992d9-40b9-4d0f-f.txt	.txt	491030	76	888460	4	42161669176	9/16/26 5:32 PM	FileCreate	Archive	NotContentIndexed	14353464	C:\Users\CyberSamurai\Desktop\Si							
116128	WER.2f6992d9-40b9-4d0f-f.txt	.txt	491030	76	888460	4	42161669316	9/16/26 5:32 PM	DataExtend	FileCreate	Archive	NotContentIndexed	14353624	C:\Users\CyberSamurai\Desktop\Si						
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117242	WER.334631e3-clae-43f3-a.txt	.txt	433784	77	888460	4	42163217584	9/16/26 5:47 PM	DataExtend	FileCreate	Archive	NotContentIndexed	15901872	C:\Users\CyberSamurai\Desktop\Si						
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117942	client_report.txt	.txt	40463	77	5704	105	42186098184	9/16/26 9:10 PM	DataExtend	FileCreate	Archive	NotContentIndexed	38782472	C:\Users\CyberSamurai\Desktop\Si						
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118270	WER.b0d04dec-3371-41ae-f.txt	.txt	490139	78	888460	4	42149274800	9/16/26 3:33 PM	DataExtend	FileCreate	Archive	NotContentIndexed	1959088	C:\Users\CyberSamurai\Desktop\Si						
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122502	WER.32230ba4-bc30-4f8e-f.txt	.txt	40958	81	888460	4	42180517888	9/16/26 8:10 PM	FileCreate	Archive	NotContentIndexed	33202176	C:\Users\CyberSamurai\Desktop\Si							
122503	WER.32230ba4-bc30-4f8e-f.txt	.txt	40958	81	888460	4	42180518048	9/16/26 8:10 PM	DataExtend	FileCreate	Archive	NotContentIndexed	33202336	C:\Users\CyberSamurai\Desktop\Si						
122506	WER.32230ba4-bc30-4f8e-f.txt	.txt	40958	81	888460	4	42180518208	9/16/26 8:10 PM	DataExtend	FileCreate	Archive	NotContentIndexed	33202496	C:\Users\CyberSamurai\Desktop\Si						
123118	WER.13984905-006a-4d6a-a.txt	.txt	251568	83	888460	4	42186870996	9/16/26 9:13 PM	FileCreate	Archive	NotContentIndexed	39554384	C:\Users\CyberSamurai\Desktop\Si							

Learned

1172 of 372204 records found

Copy Accessibility Unavailable

Average: 602/219041

Count: 36

Sum: 1.26675E+11

100%

100%

10:09 PM

9/16/2026

Parsed \$LogFile

\$LogFile captures the NTFS transaction associated with the creation of the file, corroborating the journal and MFT records.

ConsoleHost History

This artifact does not prove the file exists on its own; it corroborates that the command responsible for creating the file was executed.

```

C:\Users\CyberSamurai>ipconfig
ipconfig
net --install
net
dir
ls
cls
Get-Content "C:\Users\CyberSamurai\Downloads\basic-text.pdf" -Stream Zone.Identifier
Get-Content "C:\Users\CyberSamurai\Downloads\basic-text.pdf" -Stream Zone.Identifier
Get-Content "C:\Users\CyberSamurai\Downloads\basic-text.pdf" -Stream Zone.Identifier
Get-Content "C:\Users\CyberSamurai\Pictures\IMG_4899.JPG" -Stream Zone.Identifier
Get-Content "C:\Users\CyberSamurai\Desktop\UEFIcmd.runtimeconfig.json" -Stream Zone.Identifier
Get-Content "C:\Users\CyberSamurai\Desktop\I386_PAKSED.csv" -Stream Zone.Identifier
Get-Item "C:\Users\CyberSamurai\Desktop\UEFIcmd.runtimeconfig.json" -Stream *

cd ..
cd .\Users\CyberSamurai\Desktop\
sir
dir
ls
cls
echo Confidential report_client_report
cls
echo Confidential report_client_report
cls
echo Confidential > report_client_report
echo Confidential report > client_report.txt
delete client_report.txt
cls
cd ..
cd .\Users\CyberSamurai\Desktop\
echo Confidential report > client_report
cls
echo Confidential report > client_report.txt

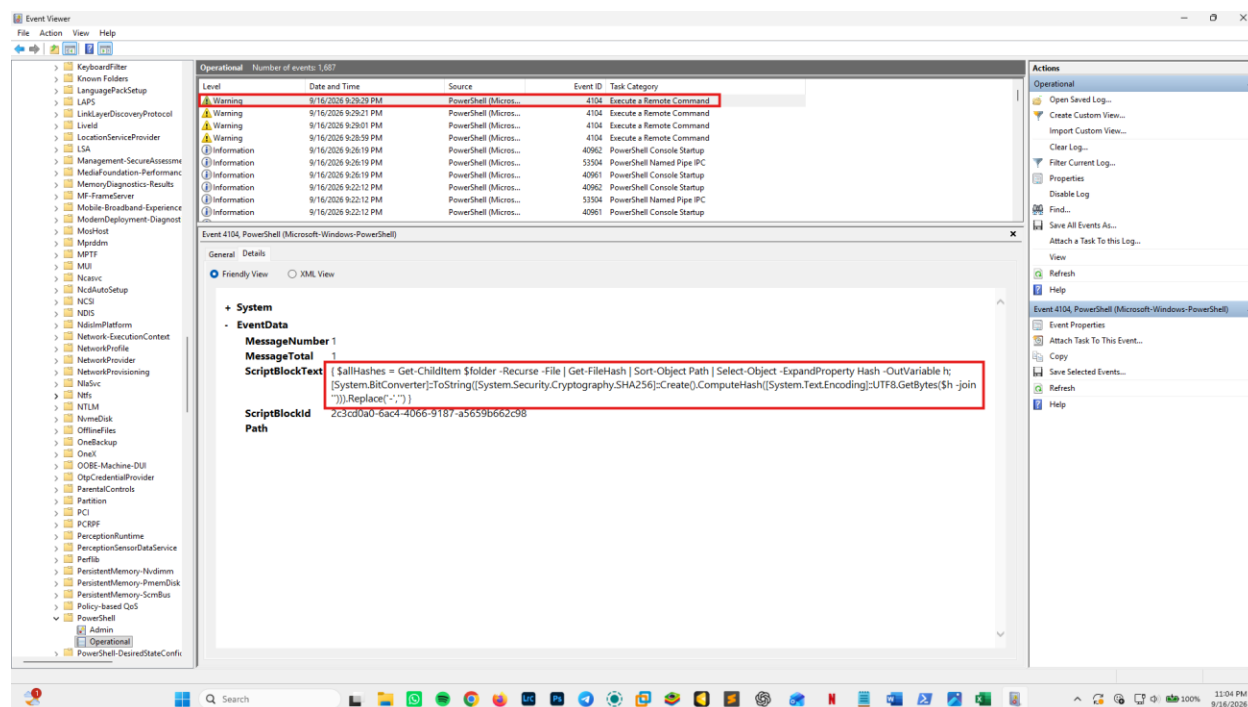
```

Step 7. Event Viewer

PowerShell Operational logs were examined during the investigation.

No event corresponding directly to the file creation timestamp was present. The available PowerShell events instead recorded activity related to the subsequent hash calculation performed during evidence processing, approximately nine minutes after the file was created.

This absence does not weaken the conclusion, as the creation event is independently corroborated by ConsoleHost history and multiple NTFS artifacts.



Conclusion

This investigation demonstrates that a file created entirely through the **Command Line Interface** leaves a distinct and corroborating chain of forensic evidence.

Rather than relying on a single artifact, the examination correlates **ConsoleHost_history.txt**, **\$MFT**, **\$UsnJrnl**, and **\$LogFile** to reconstruct the birth of client_report.txt. Together, these artifacts independently describe the same creation event and provide a defensible forensic narrative of CLI-based file creation.

As part of the broader forensic lifecycle project, this report documents the **CLI workflow**, while the corresponding GUI investigation examines the same stage through Explorer-based user

interaction. Both workflows ultimately converge on the same physical NTFS evidence, demonstrating how different user actions produce corroborating forensic artifacts.